



Unit Code:	BIT 02105.
Unit Title:	DATA STRUCTURES AND ALGORITHMS
Program(s):	BIT.
Lecturer Name:	George Wainaina.
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Consultation time:	Wednesday 2:00PM – 6:00PM

UNIT DESCRIPTION/ OBJECTIVES OF THE UNIT

The purpose of this course is to provide the learner with a strong foundation in algorithms and data structures in preparation for jobs in industry or for more advanced courses. Students will be taught on how to select and design data structures and algorithms that are appropriate for problems that they might encounter.

EXPECTED LEARNING OUTCOMES

Upon successful completion of this course, students should be able to:

- i). Describe an abstract data type (ADT) and describe its role in program design
- ii). Implement abstract data structures
- iii). Elaborate on contexts where their use is appropriate
- iv). Use recursion as an algorithm design technique in appropriate contexts
- v). Carry out the analysis of a range of algorithms in terms of algorithm analysis and express algorithm complexity using the O notation
- vi). Implement standard searching and sorting algorithms

PREREQUISITE:

Structured Programming and Algorithms.

COURSE SYLLABUS AND SCHEDULE

Week	Topic	Sub-Topic
1	Introduction and Definition of terms.	• Programming strategies • Objects and ADTS • Objects and Destructors • Data Structures • Methods • Pre and Post Conditions
2	Abstract Data Types:	• Arrays. • Queues
3	Abstract Data Types:	• Linked Lists. • Stacks.
4	Trees:	• Binary trees and binary search trees. • Tree traversals. • AVL trees, B-trees;
5	Algorithm Analysis:	• Recursion • Big-O notations. • Running time calculation. • Complexity Analysis
6	Hashing:	• Hash function. • Separate chaining. • Open addressing. • Rehashing;
7	Priority Queues (Heaps):	• Binary heaps. • Applications of priority queues. • D-heaps;
8	Sorting:	• Bubble sort. • Insertion sort. • Selection sort. • Shell sort.
9	Sorting:	• Heap sort. • Merge sort. • Quick sort. • Bucket sort. • External sorting.
10	Searching:	• Binary. • Linear. • Fibonacci, Binomial, coefficients, optimal and binary searches.
11	Graph Algorithms:	• Shortest-path algorithms. • Minimum-span tree. • Dijkstra's Algorithm
12	Revision.	
13	Revision.	

14	Final examination.	
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Mode of Delivery

The course unit will be delivered through face to face.

Teaching Methodology

Lectures, Presentations, Case studies, Lab Practical, Library Research, Simulations.

Instructional Materials and Equipment

Course texts, Handouts, Presentation slides, Computer software and hardware, Virtual Labs, Simulators, LMS, Video Conferencing.

Course Assessment

This course will be assessed as follows:

Continuous Assessment Test 1 15%
Continuous Assessment Test 2 15%
Assignment1 5%
Assignment2 5%
Assignment2 10%

Core Reading Materials for the Course

1. Lafore, R., (2017). *Data Structures and Algorithms in Java*. Pearson Education.
2. Cormen, T., Leiserson, C., Rivest R., Clifford S., (2019). *Introduction to algorithms* (2nd Ed.) The MIT Press.

Recommended Reference Materials

1. Michael T. G., Roberto T., & Mount D. M. (2011). *Data Structures and Algorithms in C++* (2nd Edition.) Willy.
2. Drozdek, A., (2005). *Data Structures and Algorithms in Java*, (2nd Ed.) Course Technology, a division of Thomson Learning, Inc.
3. Cormen, T., Leiserson, C., Rivest R., Clifford S., (2019). *Introduction to algorithms* (2nd Ed.) The MIT Press.
4. Storer, J.A., (2002). *An Introduction to Data Structures and Algorithms*. Springer.

Course Journal

1. International Journal of Data Structures ISSN: 0196-6774
2. Journal of Algorithms, ISSN: 1090-2678
3. International Journal of Data Structures and Algorithms
4. Journal of Computer and System Sciences, ISSN: 0022-0000

E-Resources

1. de Carvalho, W.F. & Zárte, L.E. (2021). A new local causal learning algorithm applied in learning analytics. *International Journal of Information and Learning Technology*, Vol. 38 No. 1, pp. 103-115.
<https://doi.org/10.1108/IJILT-04-2020-0046>
2. Liu, D., Liu, C., Zhang, C., Xu, C., Du, Z. & Wan, Z. (2018). Efficient hybrid algorithms to solve mixed discrete-continuous optimization problems: A comparative study. *Engineering Computations*,